

Ryan Baker

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Experience

Database Software Engineer Intern

Oracle

May 2026 – Aug 2026

Redwood Shores, CA

- ▶ Improving performance monitoring infrastructure within the Oracle Database team, focusing on AWR snapshots.
- Designed an intelligent timeout policy for table flushing in the AWR snapshot process to improve reliability and performance.
- Built internal tooling and scripts to streamline debugging, testing, and performance analysis workflows.

Course Instructor | Intro to Modern C++

Purdue University

Jan 2025 – Dec 2025

West Lafayette, IN

- ▶ Developed and taught a Modern C++ course at Purdue on behalf of IEEE, certifying successful students.
- Managed student evaluations, course logistics, and issued *Purdue College of Engineering* certificates to passing students.

Embedded Software Engineer Intern

Matrix Design Group

May 2025 – Aug 2025

Newburgh, IN

- ▶ Designed a lightweight logging system for resource-constrained platforms, optimizing for memory footprint.
- Improved CI pipelines by integrating automated testing and static analysis tooling, reducing manual overhead.
- Collaborated with senior engineers to refine architecture and validate reliability requirements on safety-critical deployments.

Conference Speaker

Core C++ Group

Nov 2024

Tel Aviv, Israel

- ▶ Presented “*Messing with Floating Point*”: a talk on floating-point use in safety- and performance-critical systems.

Teaching Assistant | Linux & Bash

Purdue University

Aug 2024 – Dec 2024

West Lafayette, IN

- ▶ Assisted with course content development; helped students complete assignments on Linux and bash scripting.

Education

M.S.E. Electrical & Computer Engineering, Purdue University

Concentration in Artificial Intelligence

Aug 2026 – May 2027

B.S.E. Computer Engineering, Purdue University

Concentration in Computer Systems | GPA: 3.5 of 4.0

Aug 2023 – May 2026

Projects

Trillim AI | trillim.com

Dec 2025 – Present

- Built a local AI inference platform achieving faster inference than Microsoft’s BitNet and llama.cpp for low-bit models.
- Shipped and maintain a Python SDK for AI developer workflows, exposing high-performance C++ inference kernels.

GitVista | gitvista.io

Oct 2025 – Present

- Developed an interactive real-time Git repository visualizer for exploring branch history and commit structure.

YouTube Channel | youtube.com/@rybkr

Feb 2024 – Present

- Produce high-quality technical programming content for 8,000+ subscribers, with 250,000+ total views.

Juicer | C++ Chess Engine

Jun 2024 – Sep 2024

- Developed a chess move generator that outperforms Stockfish (world’s foremost engine) by ~2.6x.

Skills

Languages C++, C, Go, Python, Lua, Shell, JavaScript, SQL, MATLAB

Technologies Git, Codex, Claude Code, Docker, CMake, GDB, Neovim

Frameworks Flask, Pandas, NumPy

Platforms Linux, FreeRTOS, OCI, AWS